

Mingxu Liu
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Education

- 2023–2027 **Ph.D., Computer Science**, Johns Hopkins University, Baltimore, MD
- Member of the **Biomechanical- and Image-Guided Surgical Systems (BIGSS) Lab**, Johns Hopkins University.
 - Research Assistant at the **Institute for Integrative and Innovative Research (I³R)**, University of Arkansas.
 - Research focuses on **medical robotics, medical image analysis, computer vision, image-guided intervention, and augmented reality**.
 - *Expected graduation: December 2027.*
- 2021–2022 **M.S.E., Robotics**, Johns Hopkins University, Baltimore, MD
- Member of the **BIGSS Lab**.
 - Research focused on **medical robotics and image-guided surgical systems**.
- 2016–2020 **B.Eng., Mechanical Engineering**, University of Leeds, Leeds, UK
- Research concentration in **imitation learning and autonomous driving**.
 - Developed learning-based methods for **self-driving vehicle control and decision-making**.

Research Experience

Research Experience

- 2023–Present **Graduate Research Assistant**, Johns Hopkins University, Baltimore, MD
- **Real-Time Robotic Perception & Multimodal Sensing:** Developing a real-time multimodal perception framework for continuum surgical robots, integrating fluoroscopic imaging with external tracking and robot-state information; demonstrated **sub-millimeter, low-latency 3D shape sensing**.
 - **C-Arm Kinematics & Positioning Guidance:** Developed and validated 6-DoF C-arm positioning guidance using geometric modeling, calibration, tracking, and real-time pose estimation to support efficient acquisition of target fluoroscopic views.
 - **Fluoroscopic AR Navigation:** Led development of a real-time image-guided system integrating C-arm calibration, HoloLens tracking, CUDA-accelerated DRR generation, X-ray augmentation, and ZeroMQ communication; **reduced X-ray usage by 62.3%** and increased breach-free screw diameter from **5.9 mm to 12.95 mm** in surgeon evaluation. [\[Preprint\]](#)
 - **MRI-to-Fluoroscopy Registration:** Co-led an end-to-end framework combining U-Net segmentation, cGAN-based MRI-to-CT synthesis, DRR generation, and multi-view 2D–3D reg-

istration; achieved 2.4 ± 1.0 mm translation and $1.6 \pm 0.8^\circ$ rotation error in cadaveric evaluation. [Paper]

- **MRI Segmentation & Osteonecrosis Localization:** Developed 2D/3D U-Net models for T1-weighted MRI segmentation of the femur, pelvis, and osteonecrotic lesions, achieving **0.93 Dice** for femur and **0.91 Dice** for pelvis; integrated segmentation, registration, and DRRs for intraoperative localization of pathology not visible under fluoroscopy. [Paper]
- **Multi-Device Tracking & Pose Optimization:** Developed a device-agnostic tracking framework integrating HoloLens 2 and optical trackers through dynamic scene graphs and pose-graph optimization; reduced translational error by approximately **80%** and tracking-loss rate by **30%** versus the best single-device baseline. [Paper]
- **AR Arthroscopy Reconstruction & Guidance:** Investigated augmented reality for enhancing arthroscopic visualization and surgical guidance through articular-surface reconstruction, geometric measurement, and image-based navigation. [Paper]
- **Mixed-Reality K-Wire Navigation:** Contributed to the development and evaluation of *Straight-Track*, a mixed-reality navigation system for percutaneous K-wire placement, combining instrument tracking with real-time 3D visualization to improve trajectory guidance in complex anatomy. [Paper]
- **Multimodal Pelvic Fracture AI Benchmark:** Built datasets and evaluation infrastructure for CT/X-ray pelvic-fracture segmentation and co-organized the MICCAI PENGWIN challenges, spanning **48,600 simulated X-rays** and later expanding to **500 CT cases and 340 participants worldwide**. [Paper]

2026–Present **Research Assistant**, Institute for Integrative and Innovative Research (I³R), University of Arkansas, Fayetteville, AR

- **Single-View X-ray Depth Estimation:** Developing learning-based methods for estimating anatomical depth from single fluoroscopic images using a combination of synthetic and real imaging data for image-guided intervention.
- **Image-Based Surgical Navigation:** Leading development of image-based navigation methods for aligning preoperative anatomical information and surgical plans with intraoperative fluoroscopy to support real-time localization and procedural guidance.
- **Continuum-Robot Teleoperation:** Developing and evaluating image-guided teleoperation for steerable continuum robots in orthopedic procedures, integrating real-time perception, navigation, and operator visualization in cadaveric experiments.
- **Continuum-Robot Control & Automation:** Developing integrated sensing and control methods for steerable continuum-robot systems toward increasingly automated execution of image-guided surgical tasks.

Other Experience

2022–2023 **Computer Vision Engineer**, PediaMetrix, Rockville, MD

- **RGB-D 3D Reconstruction:** Developed an end-to-end infant-head reconstruction pipeline combining deep-learning segmentation, point-cloud generation, multi-frame registration, bundle adjustment, and geometric processing for robust reconstruction under motion and depth artifacts.
- **3D Cranial Morphometrics:** Designed automated geometric measurement algorithms using

camera geometry, facial landmarks, smoothing, and outlier rejection; achieved **5.0 mm MAE (1.1%)** in head circumference and **1.6-point MAE (1.9%)** in cranial index.

- **Clinical CV/ML Validation:** Validated and refined a 2D cranial-shape measurement pipeline across **445 cases**, achieving **0.94–0.95 Spearman correlation** with 3D reference measurements; results presented at Pediatric Academic Societies 2023. [PAS 2023]

Publications

 Google Scholar

† → Equal contribution

Journal Articles

- J1. Sang, Y. *et al.* Benchmark of Segmentation Techniques for Pelvic Fracture in CT and X-Ray: Summary of the PENGWIN 2024 Challenge. *IEEE Transactions on Medical Imaging* **45**, 2212–2228. <https://doi.org/10.1109/TMI.2025.3650126> (2026).
- J2. Zou, X., Liu, M., Jones, T. T., Millan, B., Gurram, S., Pinto, P. A., Freidlin, R. & Martin-Gomez, A. Visual Cue Interactions in AR-Guided Needle Insertion: A Prostate Biopsy-Inspired Phantom Study. *IEEE Transactions on Visualization and Computer Graphics*. arXiv: 2608.27620. <https://arxiv.org/abs/2608.27620> (2026).
- J3. Ku, P.-C., Liu, M., Grupp, R., Harris, A., Oni, J. K., Mears, S. C., Martin-Gomez, A. & Armand, M. End-to-End 2D/3D Registration from Pre-Operative MRI to Intra-Operative Fluoroscopy for Orthopedic Procedures. *International Journal of Computer Assisted Radiology and Surgery* **20**. Ping-Cheng Ku and Mingxu Liu contributed equally., 2355–2366. <https://doi.org/10.1007/s11548-025-03426-w> (2025).
- J4. Shu, H., Liu, M., Seenivasan, L., Gu, S., Ku, P.-C., Knopf, J., Taylor, R. H. & Unberath, M. Seamless Augmented Reality Integration in Arthroscopy: A Pipeline for Articular Reconstruction and Guidance. *Healthcare Technology Letters* **12**, e12119. <https://doi.org/10.1049/htl2.12119> (2025).
- J5. Zhang, H., Killeen, B. D., Ku, Y.-C., Seenivasan, L., Zhao, Y., Liu, M., Yang, Y., Gu, S., Martin-Gomez, A., Taylor, R. H., Osgood, G. & Unberath, M. StraightTrack: Towards Mixed Reality Navigation System for Percutaneous K-Wire Insertion. *Healthcare Technology Letters* **11**, 355–364. <https://doi.org/10.1049/htl2.12103> (2024).
- J6. Liu, M., Martin-Gomez, A., Oni, J. K., Mears, S. C. & Armand, M. Towards Visualizing Early-Stage Osteonecrosis Using Intraoperative Imaging Modalities. *Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization* **11**, 1234–1242. <https://doi.org/10.1080/21681163.2022.2157329> (2023).

Peer-reviewed Conference Proceedings

- C1. Zhang, J., Liu, M., Shu, H., Liang, R., Liu, Y., Taskar, O., Kheradmand, A., Armand, M. & Martin-Gomez, A. Extend Your Horizon: A Device-Agnostic Surgical Tool Tracking Framework with Multi-View Optimization for Augmented Reality. In *2026 IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)* 678–685 (IEEE, 2026). <https://doi.org/10.1109/VR67842.2026.00087>.

- C2. Zou, X., Liu, M., Shu, H., Liang, R., Unberath, M. & Martin-Gomez, A. The Role of Mixed and Augmented Reality in Medical Visualization: Literature Review and A Context-Aware Taxonomy. In *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)* Accepted (2026). arXiv: [2608.27644](https://arxiv.org/abs/2608.27644). <https://arxiv.org/abs/2608.27644>.

Manuscripts Under Review

- W1. Liu, M., Liu, Z., Cai, R., Ku, Y.-C., Gu, S., Jain, A., Martin-Gomez, A. & Armand, M. *X-GuideAR: An Augmented Reality Framework to Mitigate Radiation Exposure during Fluoroscopic Guidance* Manuscript under review at International Journal of Computer Assisted Radiology and Surgery. 2026. arXiv: [2607.10873](https://arxiv.org/abs/2607.10873) [cs.CV]. <https://arxiv.org/abs/2607.10873>.
- W2. Shu, H., Seenivasan, L., Liu, M., Hwang, Y., Ku, Y.-C., Knopf, J., Martin-Gomez, A., Armand, M. & Unberath, M. *DualVision ArthroNav: Investigating Opportunities to Enhance Localization and Reconstruction in Image-Based Arthroscopy Navigation via External Cameras* Preprint. 2025. arXiv: [2511.10699](https://arxiv.org/abs/2511.10699) [cs.CV]. <https://arxiv.org/abs/2511.10699>.

Academic Service

2026	Organizer , PENGWIN 2026, MICCAI 2026	Apr. 2026–Oct. 2026
2026	Reviewer , IEEE International Symposium on Mixed and Augmented Reality (ISMAR)	
2024	Organizer , PENGWIN 2024, MICCAI 2024	Apr. 2024–Oct. 2024

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